Pixels Without Geometry?

Auditing the 3D Consistency of Video World Models with Structure-from-Motion

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CoRL 2026 Workshop: Do Robots Need World Models? • Austin, TX

code

A frozen first frame beats real robot footage on standard geometric consistency scores.
Raw scores reward predictions that barely move. Compare each prediction against **real footage with the same image motion** before reading anything into it.

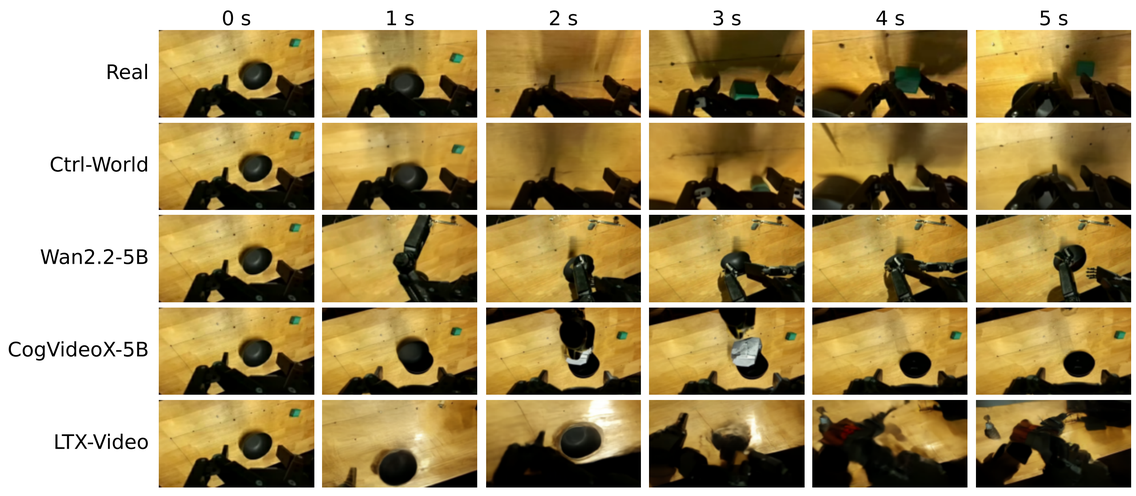
Question

Can a video model's predicted future of the robot's wrist camera be read as one rigid scene seen by a moving camera? SfM and feed-forward reconstruction are increasingly used to answer this, but a static camera gives clean matches, blur removes features without changing geometry, and non-rigid distortion can still register.

Setup

- **Data:** 60 DROID-100 wrist-camera clips, 5 s, highest image motion per episode
- **Models:** LTX-Video, CogVideoX-5B, Wan2.2-5B (first frame + instruction); Ctrl-World (action-conditioned, recorded end-effector poses)
- **Scores:** COLMAP registration, SIFT+MAGSAC++ match failure, inliers, Sampson error, VGGT depth disagreement, camera-path error
- **Controls (real footage):** non-rigid warps 1–8 px, sharpness-matched blur, slowed, unrelated frames, frozen frame, and per-clip **motion-matched** same-episode footage
- Paired bootstrap 95% CIs; 25 clips shared by all models

What the predictions look like



Ctrl-World follows the real camera motion but loses texture. Wan2.2 cuts to a third-person view, CogVideoX holds the view while objects morph, LTX distorts the scene.

Scores on the 25 shared clips

	Motion (px)	Reg. ↑	Fail ↓	VGGT ↓
Real	24.2	0.59	0.11	0.028
Slowed 2.5×	10.9	0.66	0.060	0.013
Blurred	23.7	0.76	0.15	0.029
Warp 8 px	24.6	0.70	0.15	0.045
Unrelated	118.3	0.00	0.84	0.67
LTX	10.3	0.25	0.32	0.024
CogVideoX	**2.40**	0.20	0.023	**0.007**
Wan2.2	**2.57**	0.27	0.043	**0.011**
Ctrl-World	19.7	0.26	0.36	0.031

Near-static generators move about a tenth as much as reality and get the *best* VGGT scores. Slowing real footage lowers VGGT disagreement too.

Limitations

One dataset and camera at low resolution; 25 shared clips; small open models; DROID-100 likely in Ctrl-World's training data; arm and object motion break rigidity; VGGT scores reflect its own predictions. We do not claim these scores predict planning success.

1. Raw comparisons reward uninformative output

Against the real continuation, a **frozen frame** has significantly lower match failure (-0.073), more inliers (+177) and lower VGGT disagreement (-0.031), while registering 64 points fewer frames. CogVideoX and Wan2.2 also look *better* than real footage on their matchable clips.

2. Motion-matched references shrink or remove the advantages

Model	vs.	Fail	VGGT	Reg.	n
CogVideoX	real	-0.26^*	-0.039^*	-0.29	6
	matched	-0.083^*	$\mathbf{-0.011}^*$	-0.10	6
Wan2.2	real	-0.18^*	-0.010	-0.20	6
	matched	-0.083	**+0.005**	-0.13	6
LTX	real	$+0.32^*$	-0.000	-0.41^*	15
	matched	$+0.38^*$	$+0.008$	-0.32^*	15
Ctrl-World	real	$+0.23^*$	$+0.000$	-0.39^*	22
	matched	$+0.28^*$	$\mathbf{+0.008}^*$	-0.34^*	22
Frozen frame	real	-0.073^*	-0.031^*	-0.64^*	25

Model − reference, 0.4 s pairs. *: 95% CI excludes 0.

CogVideoX's VGGT advantage shrinks by about 70%; Wan2.2's disappears. The shift from raw to matched is significant for CogVideoX, Ctrl-World and LTX. Only 6–9 clips match for the near-static models, so per-model numbers are exploratory.

3. Matching can also expose a hidden deficit

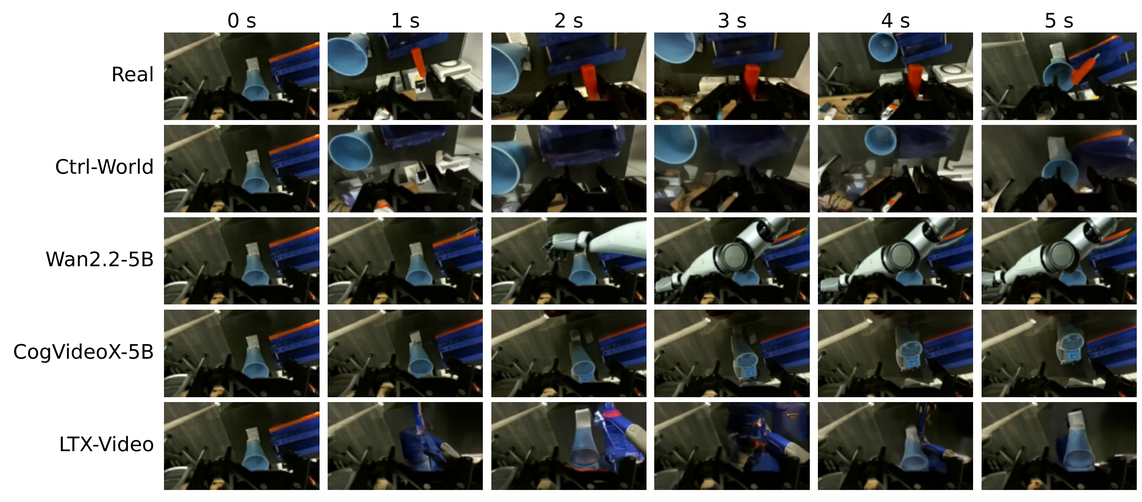
Ctrl-World moves almost as much as reality (19.7 vs. 24.2 px). Against the real continuation its VGGT disagreement shows no difference; against matched footage it is higher on the shared clips, though not significantly on all 60 clips. Registration is 33 points below real and 50 below sharpness-matched blur, so blur does not explain it.

A reporting protocol

1. Report image motion with two estimators (SIFT displacement, dense flow) and match coverage
2. Always include a **frozen-frame** control
3. Compare each clip against **motion- and appearance-matched real** footage
4. Report the scores separately, never as one number

A second scorer (StreamVGGT) agrees closely (Spearman $\rho = 0.86$), but changes the raw LTX comparison, so raw scores can also depend on the scorer.

A second clip



A clip near the median registration of Ctrl-World and LTX: Ctrl-World follows the camera's sweep; the others hold the view or distort it.

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